Math Inquirers

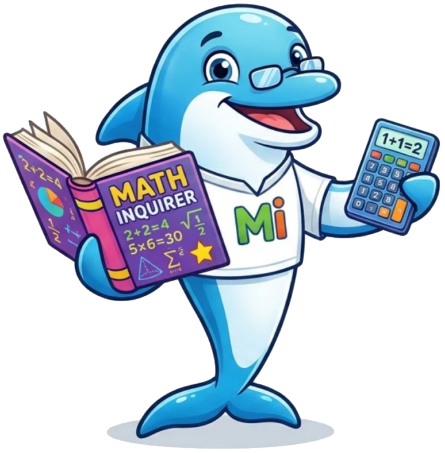
Think. Question. Prove.

Level 1 (ages 6-8) - Term 1

Workbook

Book 2

Lessons 6 to 10



My name:

Contents

Math Inquirers Level 1, Term 1 - Workbook, Book 2

Lesson 6 page 3

Lesson 7 **Take Away, Count Up** page 11
Addition And Subtraction

Lesson 8 **Balance and Swap** page 19
Addition And Subtraction: Balance

Lesson 9 **Draw and Measure** page 27
Length: Draw And Measure

Lesson 10 **Story Steps: Read It Twice** page 33
Story Steps

Kit pages (cut-outs)

Some lessons end with a kit page: tens bars, number lines, cards and so on.
Each kit page has a blank back, so you can cut or tear it out without losing any work.
Tear along the dashed line in the left margin, then cut out the pieces. Keep them in an envelope.

TENS JUMP, ONES HOP!

Name _____ Date _____



After question 3 you may choose: carry on, or try **Think harder**.

Use your kit or a drawing. A bundle of 10 is a bar in your kit. Say your answer out loud, then write it.

No.S1

These are Mei's sticks. She gets another bundle of 10 sticks. Fill in the blanks.



(a) Now Mei has _____ bundles of ten.

(b) She also has _____ single sticks.

(c) Mei has _____ sticks altogether.

No.S2

In the box, draw 3 tens as 3 tall bars and 4 single sticks. Next, draw 2 more tall bars. How many sticks are there altogether?

There are _____ sticks altogether.

No.S3

Jump forward. Fill in the blanks.

(a) 23 + 10 = _____

(b) 23 + 40 = _____

(c) 51 + 30 = _____

(d) 7 + 60 = _____

Name _____ Date _____

No.S4

Jump or hop. Fill in the blanks.

(a) 34 - 10 = _____

(b) 58 - 20 = _____

(c) 58 - 3 = _____

(d) 62 + 7 = _____

No.S5

Ali has 68 pencils. He gives 3 boxes of 10 pencils to his friend. How many pencils does Ali have left?

There are _____ pencils left.

No.S6

Ben says 45 + 20 = 47. Fill in a to c.

(a) Is Ben correct? Circle Yes or No. **Yes** **No**

(b) The right answer is _____.

(c) Ben added 2 _____ by mistake. He should add 2 _____.

No.S7

Start at 36. On a number line in the box, draw 2 jumps of ten. Where do you land?

I land on _____.

Name _____ Date _____

Use your kit or draw the jumps and hops. Then write how you know. Choose any question, in any order.

No.T1

Add 23 + 45. Fill in the blanks.

(a) Jump: 23 + _____ = _____

(b) Hop: _____ + _____ = _____

(c) So 23 + 45 = _____

No.T2

Subtract 79 - 35. Fill in the blanks.

(a) Jump back: 79 - _____ = _____

(b) Hop back: _____ - _____ = _____

(c) So 79 - 35 = _____

No.T3

Both frogs start on 12. Frog A jumps 3 tens, then hops 4. Frog B hops 4, then jumps 3 tens. Fill in a to c.

(a) Frog A lands on _____.

(b) Frog B lands on _____.

(c) Is the landing number the same for both frogs? Circle Yes or No. **Yes** **No**

No.T4

Ali starts on a number. He jumps forward 3 tens and ends on 78. Where did Ali start?

Ali started on _____.

Name _____ Date _____

Choose any question. You do not have to finish them all.

No.G1

Mei is on 7 and must get to 97. Each time she can jump 10 or jump 20. Many answers are right. Fill in a and b.

(a) Write the numbers Mei lands on in one way.

(b) Write the numbers for a different way.

No.G2

Now Mei wants the fewest jumps from 7 to 97. She can jump 10 or jump 20. Fill in a and b.

(a) The fewest number of jumps is _____.

(b) Why can Mei not do it in 4 jumps? I know because _____.

No.G3

Start at 25. Jump forward 3 tens, jump back 1 ten, then jump forward 2 tens. Then hop and land on 69. Fill in a to c.

(a) Write each number you land on before the hop.

(b) The hop is _____ ones forward.

(c) The three jumps are the same as one jump of _____ tens forward.

Name _____ Date _____

Investigation: Jump road

Try this any time. You do not have to finish.

First, try this. Start at 6. Jump in tens 5 times. Write every number you land on.

Next, try this. Start at 6 and keep jumping in tens. Will you land on 50? Write Yes or No. Then try other start numbers from 0 to 50. Which start numbers can land on 50? Write what you notice.

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

Add.

46 + 30 = _____

No.E2

Subtract.

(a) 74 - 30 = _____

(b) 74 - 3 = _____

No.E3

Mei is on 52. She jumps forward 2 tens, then hops forward 3. Fill in the blanks.

(a) After the jump she is on _____.

(b) After the hop she is on _____.

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well: .....

The hardest part was ... ☐ knowing if it is a jump or a hop ☐ taking away tens ☐ jump, then hop ☐ the number line ☐ something else

What helped me: ☐ the bars and squares ☐ drawing the jumps ☐ trying again ☐ I asked for help ☐ I was not stuck

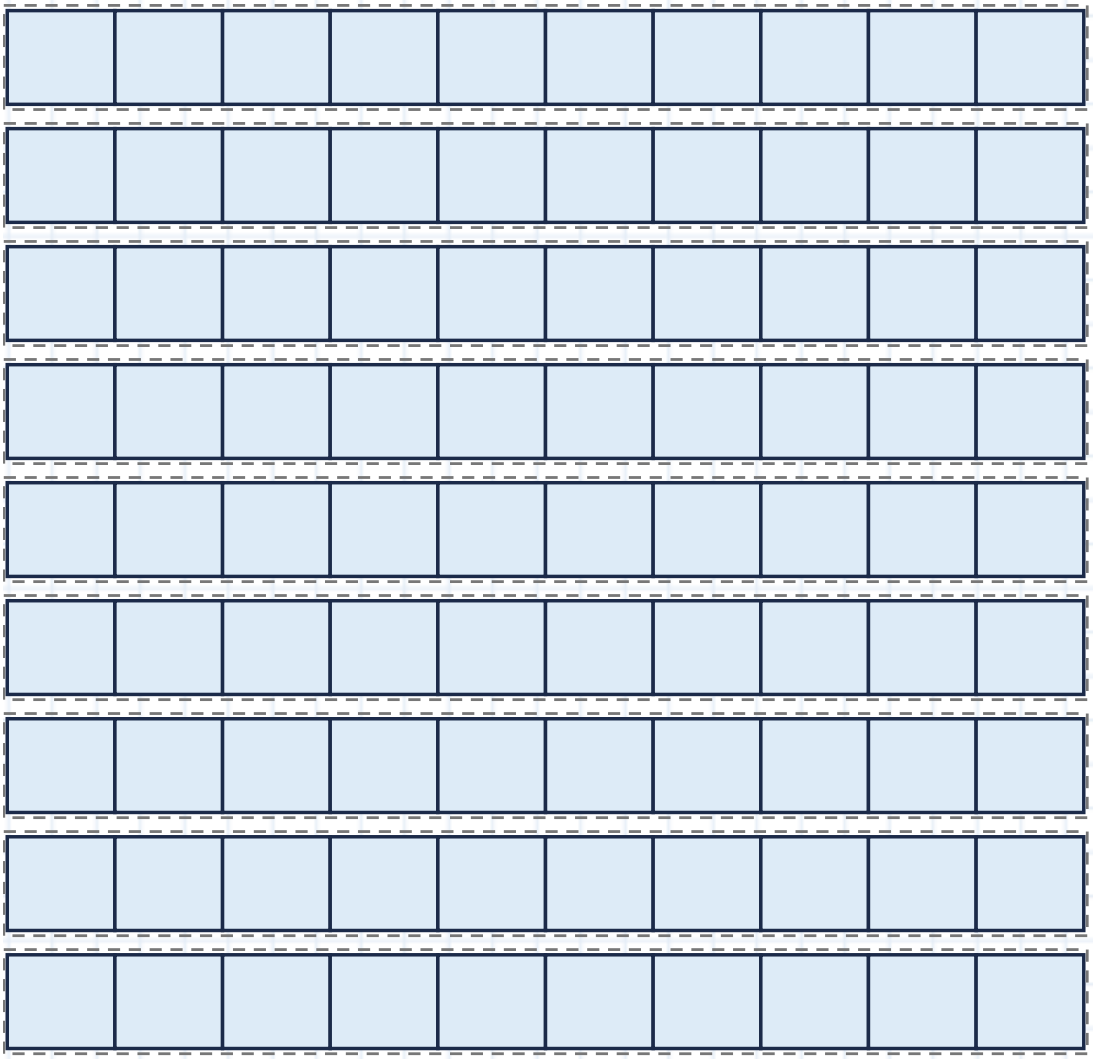
Next time I will try: ☐ drawing it ☐ using objects ☐ checking my answer ☐ asking a question ☐ I am still stuck (my grown-up will photograph this page for my tutor) ☐ I would like to pass

Name _____ Date _____

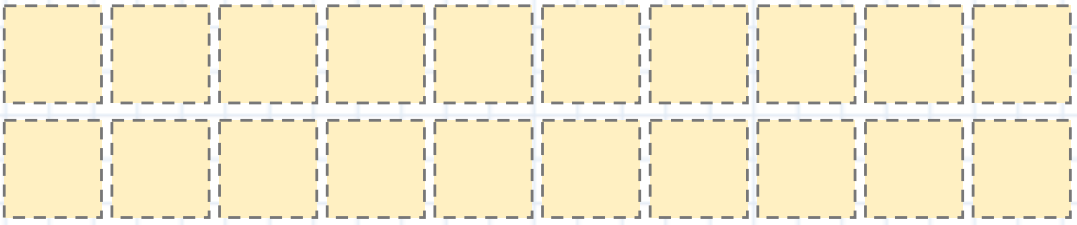
Kit page: tens bars and ones squares

A grown-up cuts these out before the lesson. A bar is a ten. A square is a one. These are the kit from the first lesson of the term, so keep them in an envelope.

Tens bars: cut out all 9 (each bar is one ten)



Ones squares: cut out all 20 (each square is one one)



TEAR OUT here (blank on the back) - then cut out the pieces

TAKE AWAY, COUNT UP!

Name _____ Date _____

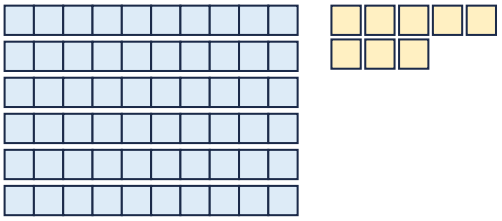


After question 3 you may choose: carry on, or try **Think harder**.

Use your kit, a drawing or your fingers. Say your answer out loud, then write it.

No.S1

Here is 68 as bars and squares. Cross out 2 bars. Then fill in the blanks.



(a) _____ tens and _____ ones are left.

(b) 68 - 20 = _____

No.S2

Subtract. Use your kit if you like.

(a) 57 - 20 = _____

(b) 57 - 4 = _____

(c) 86 - 30 = _____

(d) 86 - 5 = _____

No.S3

Find 68 - 25. Fill in the blanks.

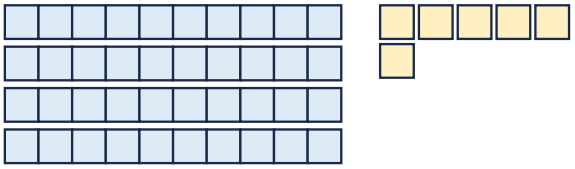
(a) 68 - 20 = _____

(b) 48 - 5 = _____

Name _____ Date _____

No.S4

Subtract. Then add to check.



(a) 46 - 12 = _____

(b) 34 + 12 = _____

No.S5

Mei has 45 stickers. She gives away 13. How many stickers are left? Draw a bar for each ten and a square for each one.

Mei has _____ stickers left.

No.S6

Kumar says 59 - 20 = 57.

(a) Is Kumar right? Circle Yes or No. **Yes** **No**

(b) 59 - 20 = _____

No.S7

A bus has 64 seats. 21 children are sitting. How many seats are empty?

_____ seats are empty.

Name _____ Date _____

Use your kit, draw a picture or count up. Then write how you know. Choose any question, in any order.

No.T1

Mei has 62 beads. Ravi has 38 beads. Find how many more beads Mei has than Ravi.



Mei has _____ more beads.

No.T2

Find 83 - 77. Show how you found it.

83 - 77 = _____

No.T3

Ali has 24 stickers and Ben has 31 stickers. Then each child gets 10 more stickers. Find how many more stickers Ben has than Ali now.

Ben has _____ more stickers.

No.T4

A book has 71 pages. Ravi has read 46 pages. How many pages are left?

_____ pages are left.

Name _____ Date _____

Choose any question. You do not have to finish them all.

No.G1

Fill in the blanks. Circle the quicker way for each.

(a) 71 - 68 = _____ **take away** **count up**

(b) 71 - 20 = _____ **take away** **count up**

(c) I know because _____.

.....

.....

No.G2

A crate has 100 eggs. Ali sells 64 of them. How many eggs are left? Find the answer two ways.

(a) By taking away: _____ eggs are left.

(b) By counting up: _____ eggs are left.

(c) Do the two answers match? Circle Yes or No. **Yes** **No**

.....

.....

No.G3

Mei says 83 - 47 and 86 - 50 have the same answer.

(a) 83 - 47 = _____

(b) 86 - 50 = _____

(c) Is Mei correct? Circle Yes or No. **Yes** **No**

(d) I know because _____.

.....

Name _____ Date _____

Investigation: When is counting up quicker?

Try this any time. You do not have to finish.

First, try this. Find 52 - 49 and 52 - 10. Try take away and count up for each one. Which way is quicker for each?

Next, try this. Write two subtractions where counting up is quicker, and two where taking away is quicker. Many answers are right. Then finish: Counting up is quicker when ___.

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

Subtract.

58 - 24 = _____

No.E2

Fill in the blanks.

(a) 57 + _____ = 62

(b) 62 - 57 = _____

No.E3

Find 90 - 87. Circle the way you used: take away or count up.

90 - 87 = _____

take away **count up**

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well:

The hardest part was ... ☐ taking away tens and ones ☐ hopping to the next ten ☐ counting up in jumps ☐ choosing a tool ☐ something else

What helped me: ☐ bars and squares ☐ drawing it ☐ trying again ☐ I asked for help ☐ I was not stuck

Next time I will try: ☐ drawing it ☐ using objects ☐ checking my answer ☐ asking a question ☐ I am still stuck (my grown-up will photograph this page for my tutor) ☐ I would like to pass

Name _____ Date _____

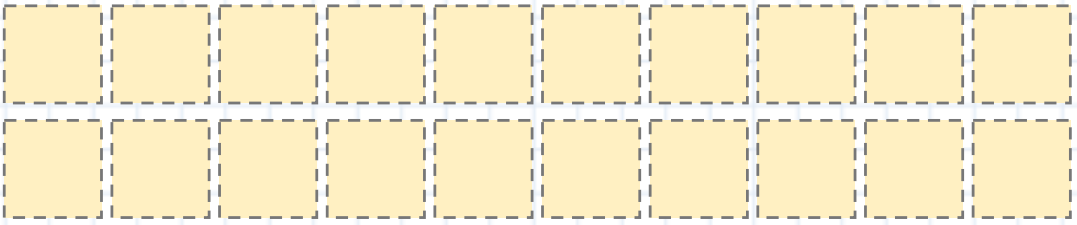
Kit page: tens bars and ones squares

A grown-up cuts these out before the lesson. A bar is a ten. A square is a one. Keep them in an envelope.

Tens bars: cut out all 9 (each bar is one ten)



Ones squares: cut out all 20 (each square is one one)



TEAR OUT here (blank on the back) - then cut out the pieces

BALANCE AND SWAP!

Name _____ Date _____



After question 3 you may choose: carry on, or try **Think harder**.

Use your plates or a drawing. Say it out loud, then write it.

No.S1

Fill in the missing numbers.

(a) 5 + 3 = 4 + ☐

(b) 6 + 4 = 5 + ☐

(c) 7 + 3 = ☐ + 7

(d) 2 + 8 = 8 + ☐

No.S2

Mei puts 6 buttons and 2 buttons on the left plate. On the right plate she puts 7 buttons and some more. Draw the right plate in the box so both plates are the same.

She adds ☐ more buttons

No.S3

Siti writes 6 + 3 = 5 + 9.

(a) Is it balanced? Circle Yes or No. **Yes** **No**

(b) Fill in the missing number: 6 + 3 = 5 + ☐

Name _____ Date _____

No.S4

7 + 5 = 12. Fill in the missing numbers.

(a) 6 + ☐ = 12

(b) 5 + ☐ = 12

(c) 4 + ☐ = 12

No.S5

Fill in the missing numbers.

(a) 20 + 30 = 40 + ☐

(b) 50 + 20 = 60 + ☐

(c) 40 + 20 = 30 + ☐

(d) 30 + 30 = 20 + ☐

No.S6

Mei has 9 red pens and 6 blue pens. Ben has 10 red pens and some blue pens. They have the same number of pens. How many blue pens does Ben have?

Ben has ☐ blue pens.

.....

No.S7

Fill in the missing number.

35 + 12 = ☐ + 2

Name _____ Date _____

Use your plates or draw a picture. Then write how you know. Choose any question, in any order.

No.T1

Mei says 47 + 18 and 45 + 20 have the same total. Ali says 45 + 20 has the greater total. Do not add.

(a) Who is right? Circle Mei or Ali.

(b) I know because _____.

.....

.....

No.T2

Do not add. Circle LEFT or RIGHT for the greater total, or circle SAME.

(a) 29 + 14 and 30 + 13: LEFT / RIGHT / SAME

(b) 53 + 20 and 50 + 23: LEFT / RIGHT / SAME

(c) 26 + 15 and 28 + 16: LEFT / RIGHT / SAME

No.T3

Fill in the missing number. Then find the total.

(a) 49 + 26 = 50 + ☐

(b) The total is ☐.

.....

No.T4

Ben writes 24 + 17 = 26 + 19.

(a) Is it balanced? Circle Yes or No. **Yes** **No**

(b) Change the 19 to ☐.

.....

Name _____ Date _____

Choose any question. You do not have to finish them all.

No.G1

Fill in the missing numbers so that all three sums have the same total.

19 + 28 = 20 + ☐ = ☐ + 30

No.G2

Fill in the blanks so both sides are the same. Write 3 different pairs. Many answers are right.

(a) ☐ + 5 = 9 + ☐

(b) ☐ + 5 = 9 + ☐

(c) ☐ + 5 = 9 + ☐

(d) I notice _____.

No.G3

Mei has 28 stickers on a page and 17 in her pocket. Ali, Ben and Kumar have the same total as Mei. Fill in the blanks.

(a) Ali has 30 stickers on his page. His pocket has ☐ stickers.

(b) Ben has 25 stickers on his page. His pocket has ☐ stickers.

(c) Kumar has 12 stickers in his pocket. His page has ☐ stickers.

Name _____ Date _____

Investigation: Keep it the same

Try this any time. You do not have to finish.

First, try this. Add 6 + 4, then 7 + 3, then 7 + 5. Which total is not 10? Then look at 6 + 4 and 7 + 3: the 6 goes up 1. What does the 4 do?

Next, try this. Pick two numbers. Add 3 to one and take 3 from the other. Then add 3 to both. Then take 2 from both. Which keep the total the same? By how much do the others change it?

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

Fill in the missing number.

9 + 5 = 10 + ☐

No.E2

Is 28 + 6 = 30 + 4 balanced?

Circle Yes or No. **Yes** **No**

No.E3

Fill in the missing number.

30 + 12 = 40 + ☐

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well: _____

The hardest part was ... ☐ knowing what = means ☐ finding the blank ☐ how the numbers moved ☐ the big numbers (Think harder) ☐ something else

What helped me: ☐ two plates and small things ☐ drawing it ☐ trying again ☐ I asked for help ☐ I was not stuck

Next time I will try: ☐ drawing it ☐ using objects ☐ checking my answer ☐ asking a question ☐ I am still stuck (my grown-up will photograph this page for my tutor)

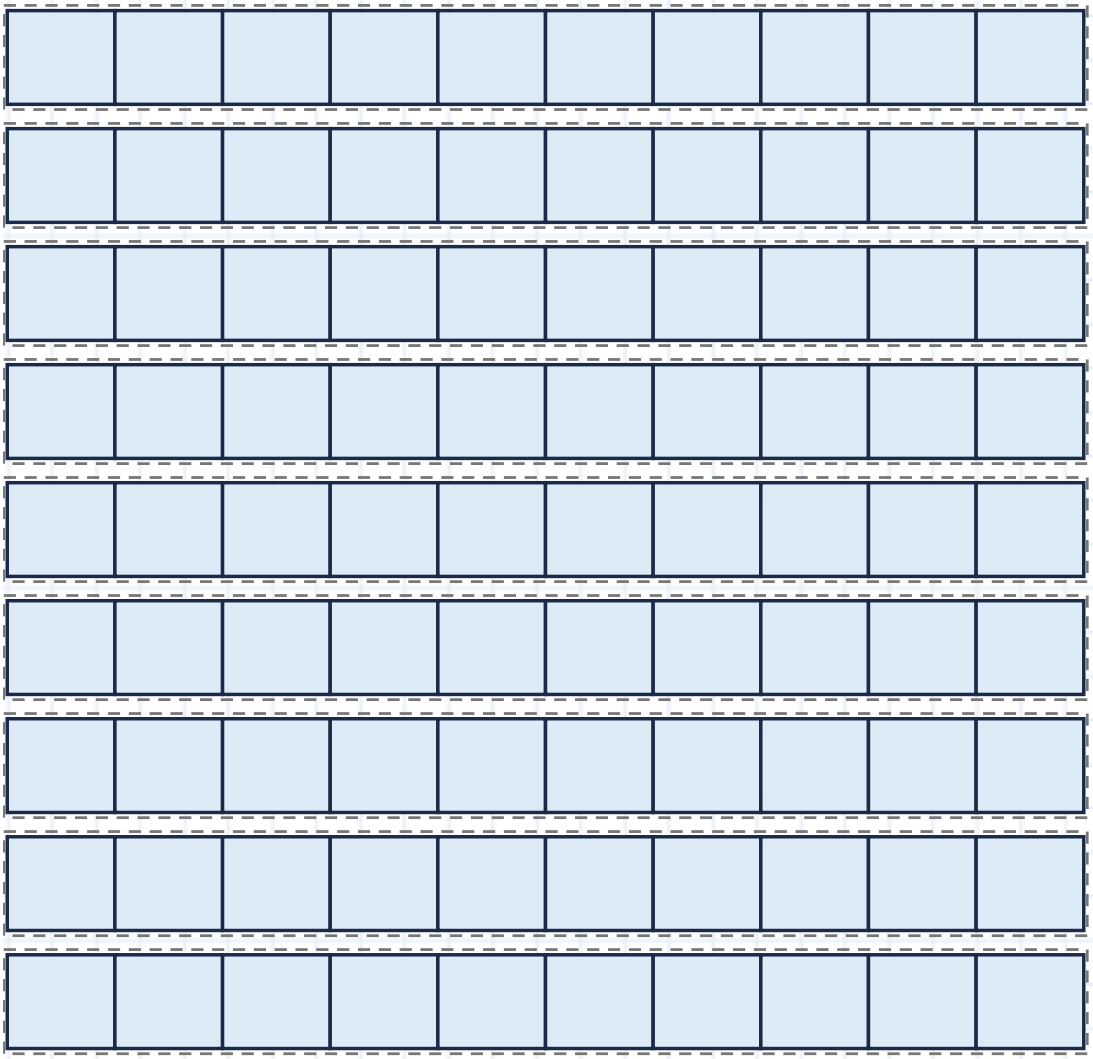
☐ I would like to pass

Name _____ Date _____

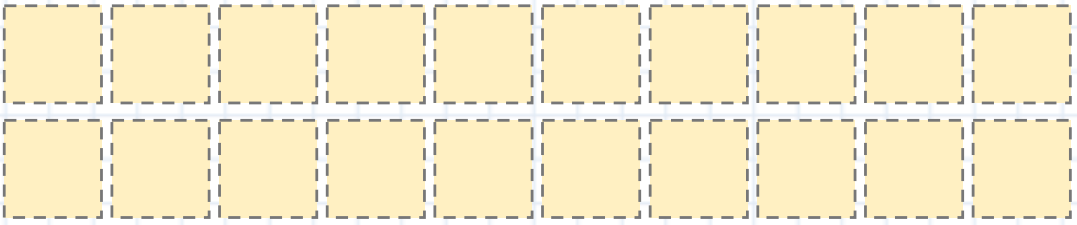
Kit page: tens bars and ones squares

A grown-up cuts these out before the lesson. A bar is a ten. A square is a one. They help with questions S5 and S7 (the tens). For the two plates, use real plates. Keep the pieces in an envelope.

Tens bars: cut out all 9 (each bar is one ten)



Ones squares: cut out all 20 (each square is one one)



TEAR OUT here (blank on the back) - then cut out the pieces

DRAW AND MEASURE!

Name _____ Date _____



After question 3 you may choose: carry on, or try **Think harder**.

Use your ruler and a pencil. Say your answer out loud, then write it.

No.S1

In the box, draw a line 7 cm long. Start at the 0 mark.

Your line ends at the ☐ mark.

No.S2

In the box, draw a line 9 cm long. Ali's line is 4 cm long.

Your line is _____ cm longer than Ali's line.

No.S3

Measure something short from your table, such as a crayon or a rubber. Write your guess first. Then measure it in whole cm.

(a) My guess: _____ cm

(b) I measured: _____ cm

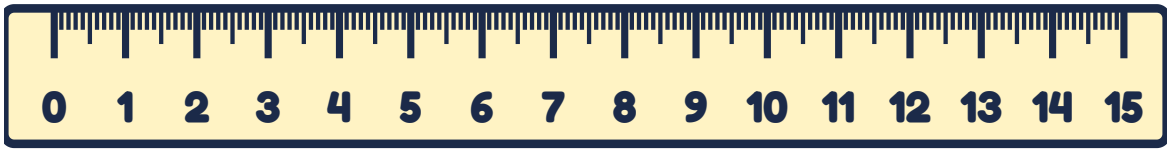
Name _____ Date _____

No.S4

Look at the ruler. Ali's pencil and Mei's pencil both start at the 0 mark.

Ali's pencil

Mei's pencil



(a) Ali's pencil is _____ cm long.

(b) Mei's pencil is _____ cm long.

(c) Ali's pencil is _____ cm longer than Mei's.

No.S5

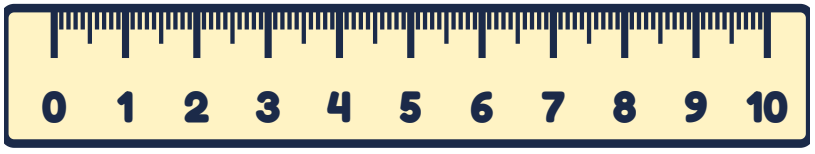
Mei's line is 5 cm long. In the box, draw a line 3 cm longer than Mei's line.

Your line is _____ cm long.

No.S6

Look at Mei's line on the ruler. Mei says it is 8 cm long.

Mei's line



(a) Is Mei correct? Circle Yes or No. **Yes** **No**

(b) Her line is _____ cm long.

Name _____ Date _____

Use your ruler and drawings to help. Choose any question, in any order.

No.T1

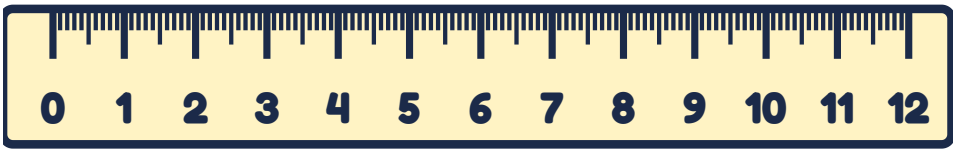
Use your ruler to draw a path with 3 straight pieces: 5 cm, 3 cm and 4 cm.

The whole path is _____ cm long.

No.T2

Look at line A on the ruler. Line B starts at the 3 mark and is 6 cm long.

Line A



(a) Line A is _____ cm long.

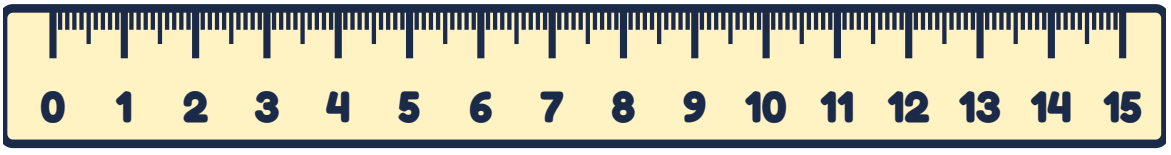
(b) Line B ends at the _____ mark.

No.T3

Look at the ruler. Both pencils start at the 0 mark. Each tip is between two marks.

Ali's pencil

Mei's pencil



(a) To the nearest cm, Ali's pencil is _____ cm long.

(b) To the nearest cm, Mei's pencil is _____ cm long.

(c) Mei's pencil is _____ cm longer than Ali's.

Name _____ Date _____

Choose any question. You do not have to finish them all.

No.G1

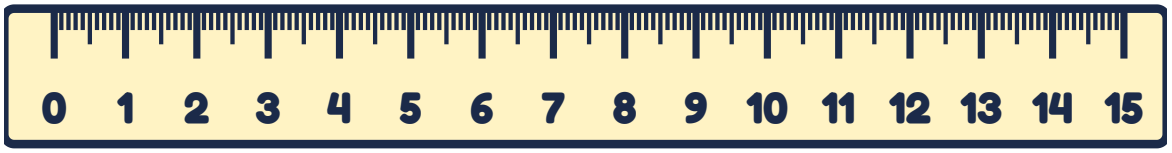
A ribbon is 30 cm long. Mei cuts all of it into 3 pieces: A, B and C. Piece A is 8 cm long. Piece B is 3 cm longer than piece A. How long is piece C?

Piece C is _____ cm long.

No.G2

A snail crawls along a long ruler from the 4 mark to the 13 mark. Then it crawls back 5 cm.

The snail's way forward



(a) The snail is now at the _____ mark.

(b) The snail has crawled _____ cm in all, counting the way back.

No.G3

Draw a path with 3 straight pieces. The pieces are all different lengths, in whole cm. The whole path is exactly 20 cm long. Many answers are right.

My 3 lengths: _____ cm, _____ cm and _____ cm.

Name _____ Date _____

Investigation: Does the start mark matter?

Try this any time. You do not have to finish.

First, try this. Draw a line 6 cm long from the 2 mark. Where does it end?

Next, try this. Try 3 more start marks, 6 cm each time. Lines 1 to 3: write the start and end marks. Line 4: I notice ...

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

In the box, draw a line 6 cm long. Start at the 0 mark.

Your line ends at the ☐ mark.

No.E2

Siti's rubber ends at the 6 mark and her pencil ends at the 15 mark. Both start at the 0 mark.

The pencil is _____ cm longer than the rubber.

No.E3

A line starts at the 3 mark and ends at the 10 mark. How long is the line?

The line is _____ cm long.

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well:

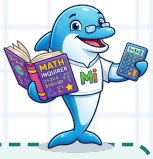
The hardest part was ... ☐ starting at the 0 mark ☐ drawing a line exactly ☐ a line that starts at another mark ☐ adding the pieces of a path ☐ something else

What helped me: ☐ my ruler ☐ drawing it ☐ trying again ☐ I asked for help ☐ I was not stuck

Next time I will try: ☐ drawing it ☐ using objects ☐ checking my answer ☐ asking a question ☐ I am still stuck (my grown-up will photograph this page for my tutor) ☐ I would like to pass

STORY STEPS: READ IT TWICE!

Name _____ Date _____



After question 3 you may choose: carry on, or try **Think harder**.

Use counters or a drawing. Read the story twice. Say your answer out loud, then write it.

No.S1

Ben has 30 cards. He gives 20 cards away. After that, he gets 6 more cards.

(a) Step 1: 30 - 20 = ☐

(b) Step 2: ☐ + 6 = ☐

(c) Ben has ☐ cards now.

No.S2

Mei has 42 stickers. She gets 7 more stickers. Then she gives 10 stickers to Ali. Write one number sentence for each step.

(a) Step 1: _____

(b) Step 2: _____

(c) Mei has _____ stickers now.

No.S3

Ravi has 25 pencils. He buys 10 more pencils. Then he loses 3 pencils. Circle the 2 number sentences that fit the story.

(a) 25 + 10 = 35

(b) 25 - 10 = 15

(c) 35 - 3 = 32

(d) 35 + 3 = 38

Ravi has _____ pencils now.

Name _____ Date _____

No.S4

Write + or - in each box.

(a) Ali has 8 shells. He finds 5 more shells. 8 ☐ 5 = 13

(b) Siti has 9 sweets. She eats 3 sweets. 9 ☐ 3 = 6

(c) Mei has 10 stickers. She gives away 4 stickers. 10 ☐ 4 = 6

(d) There are 12 children in a line. 6 more children join the line. 12 ☐ 6 = 18

No.S5

Ali has 23 marbles. Ben has 10 more than Ali. How many marbles do they have altogether? Use counters if you like.

They have _____ marbles altogether.

No.S6

A bus has 30 children. 10 children get off. Then 4 children get on. Siti writes 30 + 10 = 40 for Step 1.

(a) Is Siti right? Circle Yes or No. **Yes** **No**

(b) The bus now has _____ children.

Name _____ Date _____

Use counters or draw a picture to help. Choose any question, in any order. Pages 3 to 5 can be finished at home.

No.T1

Ravi has 15 marbles. He has 3 more marbles than Mei.

(a) Mei has _____ marbles.

(b) They have _____ marbles altogether.

No.T2

Mei is in a line to buy lunch. There are 6 children in front of her and 4 children behind her. How many children are in the line?

There are _____ children in the line.

No.T3

Ben has 12 marbles. Mei has 5 more marbles than Ben. Siti has 3 fewer marbles than Mei. How many marbles does Siti have?

Siti has _____ marbles.

Name _____ Date _____

Choose any question. Write one number sentence for each step. You do not have to finish them all.

No.G1

A shop has 60 pencils. It sells 24 pencils on Monday and 15 pencils on Tuesday. On Wednesday, a box of 30 pencils comes in. Write one number sentence for each step.

(a) Step 1: _____

(b) Step 2: _____

(c) Step 3: _____

(d) The shop has _____ pencils now.

No.G2

Step 1: 25 + 10 = 35. Step 2: 35 - 8 = 27. Write a story that fits both steps. End your story with a question. Many stories are right.

.....

.....

.....

.....

No.G3

Mei, Ali and Ben have 70 shells altogether. Mei has 25 shells. Ali has 10 more shells than Mei. Write one number sentence for each step. How many shells does Ben have?

(a) Step 1: _____

(b) Step 2: _____

(c) Step 3: _____

(d) Ben has _____ shells.

Name _____ Date _____

Investigation: Does the order matter?

Try this any time. You do not have to finish.

First, try this. Start at 20. Add 10, then take away 5. Start again at 20. Take away 5 first, then add 10. Is the answer the same?

Next, try this. Pick your own numbers. Try both orders. Is the answer always the same? When can one order not work?

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

A bus has 25 children. 10 children get off. Then 3 children get on. How many children are on the bus now?

The bus now has _____ children.

No.E2

Ben has 6 stickers. Mei has 5 more stickers than Ben. How many stickers does Mei have?

Mei has _____ stickers.

No.E3

Siti has 20 beads. Ali has 7 fewer beads than Siti. How many beads does Ali have?

Ali has _____ beads.

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well: .....

The hardest part was ... ☐ reading the story twice ☐ add or take away ☐ more than, fewer than ☐ starting Step 2 from Step 1 ☐ something else

What helped me: ☐ counters ☐ drawing it ☐ reading twice ☐ I asked for help ☐ I was not stuck

Next time I will try: ☐ drawing it ☐ counters ☐ checking ☐ asking ☐ I am still stuck (my grown-up will photograph this page for my tutor) ☐ I would like to pass

Name _____ Date _____

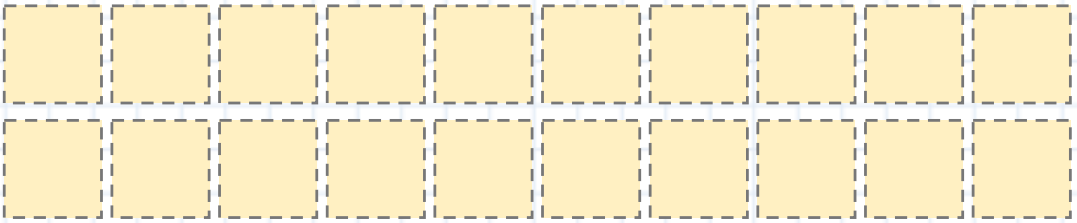
Kit page: tens bars and ones squares

Cut these out first. The squares make good counters. A bar is a ten.

Tens bars: cut out all 9 (each bar is one ten)



Ones squares: cut out all 20 (each square is one one)



TEAR OUT here (blank on the back) - then cut out the pieces